NETWORKED CONTROL

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NAME/FAMILY NAME……SOLUTIONS……………………………

PERSONAL CODE……………………………………………………

SIGNATURE……………………………………………………………

DATE……………………………………………………………………

Notes:

- The solutions must be written in the pages of this leaflet.
- At the end of the exam only this file is accepted, **without any additional page**.
- No notes or books are accepted, neither in paper nor electronic form.
- Clarity and precision will be matter of evaluation.

EXERCIZE 1

$$\dot{x} = 2x + 4u$$

A) a) $\lambda = 2 > 0$ UNSTABLE

MODE : e^{2t}

b) IF WE SET $u = kx \Rightarrow \dot{x} = (2+4k)x$

CONDITION FOR CLOSED-LOOP ASYMPTOTIC STABILITY:

$$2 + 4k < 0 \iff \boxed{k < -\frac{1}{2}}$$

B) $h = \log 2$

$x_{k+1} = f x_k + g u_k$ $\qquad f = e^{ah} = e^{2\log 2} = e^{\log 2^2} = 4$

$$g = \frac{b}{a}\left(e^{ah} - 1\right) = 2(4-1) = 6$$

a) $\lambda = f = 4 > 1 \Rightarrow$ UNSTABLE

MODE : $\lambda^k = 4^k$

b) $x_{k+1} = 4x_k + 6u_k$, WE SET $u_k = kx_k$

$= (4+6k)x_k$

$$\begin{cases} 4 + 6k < 1 \\ 4 + 6k > -1 \end{cases} \qquad \begin{aligned} k &< -\frac{3}{6} = -\frac{1}{2} \\ k &> \frac{1}{6}(-5) = \frac{-5}{6} \end{aligned}$$

$$\Rightarrow \boxed{-\frac{5}{6} < k < -\frac{1}{2}}$$

C) $h = \log_e 2$ s

a) $\tau = \log 1.1$ s

We need to find the value of k such that Φ is Schur stable, where

$$\Phi = \begin{bmatrix} e^{ah} + \Gamma(h-\tau)BKC & e^{a(h-\tau)}\Gamma(\tau)BK \\ C & 0 \end{bmatrix}$$

We compute that:

- $e^{ah} = f = 4$
- $e^{a\tau} = e^{\log 1.1^2} = 1.21$
- $\Gamma(\tau) = \frac{1}{2}\left(1.21 - 1\right) = \frac{0.21}{2}$
- $e^{a(h-\tau)^2} = \frac{4}{1.21}$
- $\Gamma(h-\tau) = \frac{1}{2}\left(\frac{4}{1.21} - 1\right) = \frac{1}{2}\,\frac{4-1.21}{1.21} = \frac{1}{2}\left(\frac{2.79}{1.21}\right)$

Therefore:

$$\Phi = \begin{bmatrix} 4 + \frac{4k}{2}\left(\frac{2.79}{1.21}\right) & \frac{4k\cdot 4}{1.21}\cdot\frac{0.21}{2} \\ 1 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 4 + 2k\left(\frac{2.79}{1.21}\right) & 8k\left(\frac{0.21}{1.21}\right) \\ 1 & 0 \end{bmatrix}$$

The characteristic polynomial of Φ is

$$p(\lambda) = \det\begin{bmatrix} \lambda - \left(4 + 2k\left(\frac{2.79}{1.21}\right)\right) & -8k\left(\frac{0.21}{1.21}\right) \\ -1 & \lambda \end{bmatrix}$$

$$= \lambda^2 - \left(4 + 2k\left(\frac{2.79}{1.21}\right)\right)\lambda - 8k\left(\frac{0.21}{1.21}\right)$$

We use the Jury criterion, i.e., Φ is Schur stable if and only if, at the same time,

(i) $$-1 < 8k\left(\frac{0.21}{1.21}\right) < 1$$

(ii) $$-1 + 8k\left(\frac{0.21}{1.21}\right) < 4 + 2k\left(\frac{2.79}{1.21}\right) < 1 - 8k\left(\frac{0.21}{1.21}\right)$$

(i) IS EQUIVALENT TO:

$$-0.72 = -\frac{1.21}{8(0.21)} < K < \frac{1.21}{8(0.21)}$$

(ii) IS EQUIVALENT TO:

$$\begin{cases} 2k\left(\frac{2.79}{1.21}\right) - 8k\left(\frac{0.21}{1.21}\right) > -5 & \text{(iii)} \\ 3 < -2k\left(\frac{2.79}{1.21}\right) - 8k\left(\frac{0.21}{1.21}\right) & \text{(iv)} \end{cases}$$

(iii): $\frac{(5.58 - 1.68)K}{1.21} > -5 \iff K > -1.55$

(iv): $\frac{(5.58 + 1.68)K}{1.21} < -3 \iff K < -\frac{1}{2}$

OVERALL: $\boxed{-0.72 < K < -0.5}$

b) $K = -\frac{2}{3}$

We compute that:

- $e^{ah} = 4$
- $e^{a\tau} = e^{2\tau} \Rightarrow \Gamma(\tau) = \frac{1}{a}\left(e^{a\tau} - 1\right) = \frac{1}{2}\left(e^{2\tau} - 1\right)$
- $e^{a(h-\tau)} = e^{ah} e^{-a\tau} = 4e^{-2\tau}$
- $\Gamma(h-\tau) = \frac{1}{a}\left(e^{a(h-\tau)} - 1\right) = \frac{1}{2}\left(4\underbrace{e^{-2\tau}}_{\alpha} - 1\right)$

Therefore:

- $e^{ah} + \Gamma(h-\tau)BKC = 4 + \frac{1}{2}(4\alpha - 1)\cdot 4 \cdot \left(-\frac{2}{3}\right)$
  $= 4 - \frac{4}{3}(4\alpha - 1) = 4 + \frac{4}{3} - \frac{16}{3}\alpha = \frac{16}{3} - \frac{16}{3}\alpha = \underbrace{\frac{16}{3}(1 - \alpha)}_{\beta}$
- $e^{a(h-\tau)}\Gamma(\tau)BK = 4\alpha \cdot \frac{1}{2}\left(\alpha^{-1} - 1\right)\cdot 4 \cdot \left(-\frac{2}{3}\right) = 2(1-\alpha)\cdot -\frac{8}{3}$
  $= -\frac{16}{3}(1 - \alpha) = -\beta$

$$\Rightarrow \Phi = \begin{bmatrix} \beta & -\beta \\ 1 & 0 \end{bmatrix} \Rightarrow p(\lambda) = \det \begin{bmatrix} \lambda-\beta & +\beta \\ -1 & \lambda \end{bmatrix} = \lambda^2 - \beta\lambda + \beta$$

We apply the Jury criterion. Φ is Schur stable iff

$$\begin{matrix} -1 < \beta < 1 \\ -1-\beta < \beta < 1+\beta \end{matrix} \rightarrow \begin{cases} -1 < \beta < 1 \\ \beta > -\frac{1}{2} \end{cases} \Rightarrow -\frac{1}{2} < \beta < 1$$

This is equivalent to say that

$$\begin{cases} \frac{16}{3}(1-\alpha) < 1 & 1-\alpha < \frac{3}{16} & \alpha > \frac{13}{16} \\ \frac{16}{3}(1-\alpha) > -\frac{1}{2} & 1-\alpha > -\frac{3}{32} & \alpha < \frac{35}{32} \end{cases}$$

This, in turn, means that

$$-2\tau > \ln\frac{13}{16} \iff 2\tau < \ln\frac{16}{13}$$

$$\boxed{\tau < \ln\sqrt{\frac{16}{13}} = \ln\frac{4}{\sqrt{13}}}$$

c) We set $K = -\frac{2}{3}$, $\bar{\tau} = \log_e 1.1\ s$

$$\tilde{\Phi}_1 = \begin{bmatrix} \beta(\bar{\tau}) & -\beta(\bar{\tau}) \\ 1 & 0 \end{bmatrix} \quad \text{where } \beta = 0.9256$$

$$\tilde{\Phi}_0 = \begin{bmatrix} f & gK \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 4 & -4 \\ 0 & 1 \end{bmatrix}$$

$$\bar{\bar{\Phi}} = \tilde{\Phi}_0 \tilde{\Phi}_1 = \begin{bmatrix} 4 & -4 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \beta & -\beta \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 4\beta - 4 & -4\beta \\ 1 & 0 \end{bmatrix}$$

The characteristic polynomial of $\bar{\bar{\Phi}}$ is

$$p(\lambda) = \det \begin{bmatrix} \lambda - (4\beta - 4) & 4\beta \\ -1 & \lambda \end{bmatrix} = \lambda^2 - (4\beta - 4)\lambda + 4\beta$$

$\bar{\bar{\Phi}}$ is Schur stable iff

(i) $-1 < 4\beta < 1$ $\Rightarrow -\frac{1}{4} < \beta < \frac{1}{4}$ NO! VIOLATED!

(ii) $-1 - 4\beta < 4\beta - 4 < 1 + 4\beta$

EXERCIZES 2 AND 3 : SEE THEORY